Table 1: Complete nonlinear-extension AUROC and AP results, with metric-specific coverage.

One-sample - Additive sinusoidal				
Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.701	100	0.348	100
uPU logistic	0.702	100	0.348	100
nnPU logistic	0.702	100	0.348	100
Naive AdaBoost	0.752	100	0.432	100
Naive GBM	0.754	100	0.426	100
Naive XGBoost	0.758	100	0.442	100
Naive LightGBM	0.752	100	0.428	100
Naive CatBoost	0.751	100	0.431	100
Corrected AdaBoost	0.752	100	0.432	100
Corrected GBM	0.754	100	0.426	100
Corrected XGBoost	0.758	100	0.442	100
Corrected LightGBM	0.752	100	0.428	100
Corrected CatBoost	0.751	100	0.431	100
RN AdaBoost	0.718	100	0.372	100
RN GBM	0.708	100	0.359	100
RN XGBoost	0.715	100	0.375	100
RN LightGBM	0.701	100	0.344	100
RN CatBoost	0.711	100	0.358	100
CS AdaBoost	0.752	100	0.432	100
CS GBM	0.756	100	0.439	100
CS XGBoost	0.759	100	0.444	100
CS LightGBM	0.752	100	0.432	100
CS CatBoost	0.754	100	0.438	100
AdaPU	0.723	100	0.381	100
PU gradient boosting	0.739	100	0.412	100

One-sample - Radial / concentric				
Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.501	100	0.246	100
uPU logistic	0.501	100	0.247	100
nnPU logistic	0.501	100	0.247	100
Naive AdaBoost	0.783	100	0.586	100
Naive GBM	0.784	100	0.601	100
Naive XGBoost	0.787	100	0.608	100
Naive LightGBM	0.784	100	0.597	100
Naive CatBoost	0.789	100	0.617	100
Corrected AdaBoost	0.783	100	0.586	100
Corrected GBM	0.784	100	0.601	100
Corrected XGBoost	0.787	100	0.608	100
Corrected LightGBM	0.784	100	0.597	100
Corrected CatBoost	0.789	100	0.617	100
RN AdaBoost	0.485	100	0.205	100
RN GBM	0.465	99	0.192	99
RN XGBoost	0.473	100	0.198	100
RN LightGBM	0.481	100	0.203	100
RN CatBoost	0.474	100	0.200	100
CS AdaBoost	0.784	100	0.589	100
CS GBM	0.786	100	0.606	100

One-sample - Radial / concentric

Method	AUROC	n AUROC	AP	n AP
CS XGBoost	0.788	100	0.607	100
CS LightGBM	0.783	100	0.598	100
CS CatBoost	0.789	100	0.616	100
AdaPU	0.789	100	0.612	100
PU gradient boosting	0.768	100	0.582	100

One-sample - Pure XOR interaction

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.500	100	0.220	100
uPU logistic	0.500	100	0.220	100
nnPU logistic	0.500	100	0.220	100
Naive AdaBoost	0.540	100	0.238	100
Naive GBM	0.753	100	0.444	100
Naive XGBoost	0.787	100	0.499	100
Naive LightGBM	0.772	100	0.477	100
Naive CatBoost	0.829	100	0.580	100
Corrected AdaBoost	0.540	100	0.238	100
Corrected GBM	0.753	100	0.444	100
Corrected XGBoost	0.787	100	0.499	100
Corrected LightGBM	0.772	100	0.477	100
Corrected CatBoost	0.829	100	0.580	100
RN AdaBoost	0.524	100	0.221	100
RN GBM	0.525	100	0.215	100
RN XGBoost	0.529	100	0.221	100
RN LightGBM	0.537	100	0.226	100
RN CatBoost	0.543	100	0.235	100
CS AdaBoost	0.544	100	0.242	100
CS GBM	0.763	100	0.466	100
CS XGBoost	0.789	100	0.496	100
CS LightGBM	0.776	100	0.481	100
CS CatBoost	0.829	100	0.578	100
AdaPU	0.500	100	0.202	100
PU gradient boosting	0.580	100	0.290	100

One-sample - Threshold interaction

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.501	100	0.201	100
uPU logistic	0.501	100	0.201	100
nnPU logistic	0.501	100	0.201	100
Naive AdaBoost	0.499	100	0.200	100
Naive GBM	0.537	100	0.221	100
Naive XGBoost	0.548	100	0.233	100
Naive LightGBM	0.583	100	0.260	100
Naive CatBoost	0.669	100	0.334	100
Corrected AdaBoost	0.499	100	0.200	100
Corrected GBM	0.537	100	0.221	100
Corrected XGBoost	0.548	100	0.233	100
Corrected LightGBM	0.583	100	0.260	100

One-sample - Threshold interaction

Method	AUROC	n AUROC	AP	n AP
Corrected CatBoost	0.669	100	0.334	100
RN AdaBoost	0.500	100	0.200	100
RN GBM	0.502	100	0.201	100
RN XGBoost	0.501	100	0.201	100
RN LightGBM	0.503	100	0.201	100
RN CatBoost	0.503	100	0.202	100
CS AdaBoost	0.499	100	0.200	100
CS GBM	0.545	100	0.229	100
CS XGBoost	0.550	100	0.233	100
CS LightGBM	0.583	100	0.261	100
CS CatBoost	0.679	100	0.341	100
AdaPU	0.500	100	0.201	100
PU gradient boosting	0.500	100	0.202	100

One-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.609	100	0.272	100
uPU logistic	0.622	100	0.282	100
nnPU logistic	0.622	100	0.282	100
Naive AdaBoost	0.737	100	0.399	100
Naive GBM	0.736	100	0.390	100
Naive XGBoost	0.740	100	0.410	100
Naive LightGBM	0.734	100	0.397	100
Naive CatBoost	0.734	100	0.400	100
Corrected AdaBoost	0.737	100	0.399	100
Corrected GBM	0.736	100	0.390	100
Corrected XGBoost	0.740	100	0.410	100
Corrected LightGBM	0.734	100	0.397	100
Corrected CatBoost	0.734	100	0.400	100
RN AdaBoost	0.730	100	0.390	100
RN GBM	0.725	100	0.380	100
RN XGBoost	0.725	100	0.381	100
RN LightGBM	0.718	100	0.370	100
RN CatBoost	0.726	100	0.387	100
CS AdaBoost	0.738	100	0.402	100
CS GBM	0.740	100	0.407	100
CS XGBoost	0.741	100	0.413	100
CS LightGBM	0.735	100	0.403	100
CS CatBoost	0.735	100	0.403	100
AdaPU	0.722	100	0.379	100
PU gradient boosting	0.721	100	0.374	100

One-sample - Nonlinear heterogeneous covariance

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.499	100	0.210	100
uPU logistic	0.499	100	0.210	100
nnPU logistic	0.499	100	0.210	100
Naive AdaBoost	0.564	100	0.250	100

One-sample - Nonlinear heterogeneous covariance

Method	AUROC	n AUROC	AP	n AP
Naive GBM	0.736	100	0.415	100
Naive XGBoost	0.779	100	0.501	100
Naive LightGBM	0.763	100	0.479	100
Naive CatBoost	0.819	100	0.574	100
Corrected AdaBoost	0.564	100	0.250	100
Corrected GBM	0.736	100	0.415	100
Corrected XGBoost	0.779	100	0.501	100
Corrected LightGBM	0.763	100	0.479	100
Corrected CatBoost	0.819	100	0.574	100
RN AdaBoost	0.513	100	0.209	100
RN GBM	0.519	100	0.209	100
RN XGBoost	0.521	100	0.210	100
RN LightGBM	0.526	100	0.214	100
RN CatBoost	0.527	100	0.219	100
CS AdaBoost	0.561	100	0.248	100
CS GBM	0.749	100	0.452	100
CS XGBoost	0.782	100	0.505	100
CS LightGBM	0.764	100	0.483	100
CS CatBoost	0.820	100	0.573	100
AdaPU	0.510	100	0.210	100
PU gradient boosting	0.560	100	0.275	100

Two-sample - Additive sinusoidal

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.703	100	0.347	100
uPU logistic	0.703	100	0.348	100
nnPU logistic	0.703	100	0.348	100
Naive AdaBoost	0.751	100	0.428	100
Naive GBM	0.753	100	0.422	100
Naive XGBoost	0.757	100	0.438	100
Naive LightGBM	0.750	100	0.422	100
Naive CatBoost	0.749	100	0.426	100
Corrected AdaBoost	0.751	100	0.428	100
Corrected GBM	0.753	100	0.422	100
Corrected XGBoost	0.757	100	0.438	100
Corrected LightGBM	0.750	100	0.422	100
Corrected CatBoost	0.749	100	0.426	100
RN AdaBoost	0.718	100	0.367	100
RN GBM	0.710	100	0.358	100
RN XGBoost	0.717	100	0.374	100
RN LightGBM	0.705	100	0.348	100
RN CatBoost	0.712	100	0.357	100
CS AdaBoost	0.751	100	0.428	100
CS GBM	0.755	100	0.433	100
CS XGBoost	0.757	100	0.439	100
CS LightGBM	0.750	100	0.426	100
CS CatBoost	0.753	100	0.433	100
AdaPU	0.726	100	0.382	100
PU gradient boosting	0.743	100	0.414	100

Two-sample - Radial / concentric

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.499	100	0.241	100
uPU logistic	0.499	100	0.246	100
nnPU logistic	0.499	100	0.246	100
Naive AdaBoost	0.782	100	0.584	100
Naive GBM	0.785	100	0.598	100
Naive XGBoost	0.786	100	0.604	100
Naive LightGBM	0.782	100	0.587	100
Naive CatBoost	0.789	100	0.616	100
Corrected AdaBoost	0.782	100	0.584	100
Corrected GBM	0.785	100	0.598	100
Corrected XGBoost	0.786	100	0.604	100
Corrected LightGBM	0.782	100	0.587	100
Corrected CatBoost	0.789	100	0.616	100
RN AdaBoost	0.480	99	0.201	99
RN GBM	0.467	99	0.193	99
RN XGBoost	0.472	100	0.197	100
RN LightGBM	0.477	100	0.200	100
RN CatBoost	0.473	99	0.200	99
CS AdaBoost	0.784	100	0.587	100
CS GBM	0.785	100	0.601	100
CS XGBoost	0.786	100	0.602	100
CS LightGBM	0.782	100	0.589	100
CS CatBoost	0.789	100	0.614	100
AdaPU	0.788	100	0.609	100
PU gradient boosting	0.764	100	0.563	100

Two-sample - Pure XOR interaction

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.501	100	0.218	100
uPU logistic	0.501	100	0.220	100
nnPU logistic	0.501	100	0.220	100
Naive AdaBoost	0.534	100	0.230	100
Naive GBM	0.735	100	0.413	100
Naive XGBoost	0.781	100	0.482	100
Naive LightGBM	0.768	100	0.465	100
Naive CatBoost	0.829	100	0.576	100
Corrected AdaBoost	0.534	100	0.230	100
Corrected GBM	0.735	100	0.413	100
Corrected XGBoost	0.781	100	0.482	100
Corrected LightGBM	0.768	100	0.465	100
Corrected CatBoost	0.829	100	0.576	100
RN AdaBoost	0.522	99	0.219	99
RN GBM	0.527	99	0.219	99
RN XGBoost	0.528	100	0.220	100
RN LightGBM	0.534	100	0.224	100
RN CatBoost	0.541	100	0.233	100
CS AdaBoost	0.542	100	0.238	100
CS GBM	0.744	100	0.432	100
CS XGBoost	0.784	100	0.481	100
CS LightGBM	0.769	100	0.467	100

Two-sample - Pure XOR interaction

Method	AUROC	n AUROC	AP	n AP
CS CatBoost	0.829	100	0.574	100
AdaPU	0.499	100	0.201	100
PU gradient boosting	0.581	100	0.272	100

Two-sample - Threshold interaction

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.501	100	0.202	100
uPU logistic	0.501	100	0.202	100
nnPU logistic	0.501	100	0.202	100
Naive AdaBoost	0.500	100	0.201	100
Naive GBM	0.530	100	0.218	100
Naive XGBoost	0.547	100	0.232	100
Naive LightGBM	0.571	100	0.251	100
Naive CatBoost	0.656	100	0.320	100
Corrected AdaBoost	0.500	100	0.201	100
Corrected GBM	0.530	100	0.218	100
Corrected XGBoost	0.547	100	0.232	100
Corrected LightGBM	0.571	100	0.251	100
Corrected CatBoost	0.656	100	0.320	100
RN AdaBoost	0.500	100	0.201	100
RN GBM	0.502	100	0.202	100
RN XGBoost	0.501	100	0.202	100
RN LightGBM	0.503	100	0.202	100
RN CatBoost	0.505	100	0.204	100
CS AdaBoost	0.500	100	0.201	100
CS GBM	0.537	100	0.224	100
CS XGBoost	0.547	100	0.231	100
CS LightGBM	0.574	100	0.253	100
CS CatBoost	0.669	100	0.331	100
AdaPU	0.500	100	0.201	100
PU gradient boosting	0.503	100	0.202	100

Two-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.604	100	0.267	100
uPU logistic	0.591	100	0.258	100
nnPU logistic	0.609	100	0.271	100
Naive AdaBoost	0.735	100	0.395	100
Naive GBM	0.731	100	0.377	100
Naive XGBoost	0.735	100	0.398	100
Naive LightGBM	0.728	100	0.384	100
Naive CatBoost	0.729	100	0.391	100
Corrected AdaBoost	0.735	100	0.395	100
Corrected GBM	0.731	100	0.377	100
Corrected XGBoost	0.735	100	0.398	100
Corrected LightGBM	0.728	100	0.384	100
Corrected CatBoost	0.729	100	0.391	100
RN AdaBoost	0.729	100	0.387	100

Two-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Method	AUROC	n AUROC	AP	n AP
RN GBM	0.721	100	0.372	100
RN XGBoost	0.724	100	0.376	100
RN LightGBM	0.715	100	0.364	100
RN CatBoost	0.723	100	0.380	100
CS AdaBoost	0.736	100	0.397	100
CS GBM	0.735	100	0.393	100
CS XGBoost	0.737	100	0.403	100
CS LightGBM	0.730	100	0.391	100
CS CatBoost	0.730	100	0.392	100
AdaPU	0.721	100	0.377	100
PU gradient boosting	0.724	100	0.373	100

Two-sample - Nonlinear heterogeneous covariance

Method	AUROC	n AUROC	AP	n AP
Logistic probability correction	0.501	100	0.209	100
uPU logistic	0.501	100	0.209	100
nnPU logistic	0.501	100	0.209	100
Naive AdaBoost	0.563	100	0.250	100
Naive GBM	0.727	100	0.396	100
Naive XGBoost	0.768	100	0.481	100
Naive LightGBM	0.750	100	0.456	100
Naive CatBoost	0.814	100	0.562	100
Corrected AdaBoost	0.563	100	0.250	100
Corrected GBM	0.727	100	0.396	100
Corrected XGBoost	0.768	100	0.481	100
Corrected LightGBM	0.750	100	0.456	100
Corrected CatBoost	0.814	100	0.562	100
RN AdaBoost	0.511	100	0.207	100
RN GBM	0.519	100	0.209	100
RN XGBoost	0.518	100	0.210	100
RN LightGBM	0.523	100	0.214	100
RN CatBoost	0.526	100	0.220	100
CS AdaBoost	0.564	100	0.250	100
CS GBM	0.736	100	0.430	100
CS XGBoost	0.770	100	0.484	100
CS LightGBM	0.753	100	0.461	100
CS CatBoost	0.816	100	0.563	100
AdaPU	0.511	100	0.210	100
PU gradient boosting	0.595	100	0.291	100

Unavailable final refits (zero-based repetition index)

Method	AUROC	n AUROC	AP	n AP
Radial / concentric One-sample, r004: RN GBM	–	0	–	0
Radial / concentric Two-sample, r086: RN AdaBoost	–	0	–	0
Radial / concentric Two-sample, r086: RN GBM	–	0	–	0

Unavailable final refits (zero-based repetition index)

Method	AUROC	n AUROC	AP	n AP
Radial / concentric Two-sample, r086: RN CatBoost	–	0	–	0
Pure XOR interaction Two-sample, r012: RN AdaBoost	–	0	–	0
Pure XOR interaction Two-sample, r083: RN GBM	–	0	–	0

Note. Means use available final refits and are conditional on refit success where n=99 (out of 100 requested). The six unavailable refits are retained explicitly above. Reason in every case: selected RN parameters yielded zero reliable negatives during final train+validation refit. No imputation, fallback model, or reselection is applied. Selected-model hard-classification results are not substituted for ranking results.